Exp:5
25/2/26

Sorting of Number.

AIM:

To write and Execute an assembly language Program for Sorting data in ascending and descending order using 8051 microcontroller on Keil Software.

Apparatus Required:

Personal Computer

Keil µ Vision Software.

Program:

```
MOV R7, #4
LOOP1: MOV R0, #4H
MOV R6, #04
LOOP: MOV A, @R0
INC R0
MOV 50H, @R0
CJNE A, 50H, NEXT.
SJMP DOWN
NEXT: JNC DOWN
MOV @R0, A
DEC R0
MOV @R0, 50H
DOWN: DJNZ R6, LOOP
DJNZ R7, LOOP1
END.
```

OUTPUT

Final Sorted Numbers

Address	Value
40H	05H
41H	10H
42H	20H
43H	25H
44H	35H

Result:

Thus the sorting of data is calculated using 8051 in Keil software.

8051 Descending Program.

Aim:

To write and execute an Assembly language program for sorting data in descending order using 8051 in Keil Software.

Apparatus Required.

Personal Computer.

Keil µVision Software.

Program:

```
MOV R7, #04H
LOOP1: MOV R0, #40H
MOV R6, #04H
LOOP: MOV A, @R0
INC R0
MOV 50H, @R0
CJNE A, 50H, NEXT
SJMP DOWN
NEXT: JC DOWN
MOV @R0, A
DEC R0
MOV @R0, 50H
DOWN: DJNZ R6, LOOP
DJNZ R7, LOOP1
END.
```

Result:

Thus, the sorting of data in descending order is calculated using 8051 in Keil Software.